

1. What is Data?

Data is a collection of raw facts, figures, or information that can be used for analysis or processing.

- Student marks
- Employee names¹.
- Temperatures
- Photos and videos

2. What is a Database?

A database is an organized collection of data stored electronically so that it can be easily accessed, managed, and updated.

- Example: A school database stores student details, attendance, and marks.

3. Why Do Organizations Store Data in Databases?

Organizations store data in databases because:

- Data can be stored systematically
- Information can be retrieved quickly
- Data security is improved
- Duplicate data is reduced
- Backup and recovery are possible
- It helps in analysis and decision making
- Example: Banks store customer accounts and transaction details in databases.

4. Examples of Relational Databases

Some popular relational databases are:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite
- MariaDB

5. What is a Data Warehouse?

A data warehouse is a centralized system used to store large amounts of historical data from different sources for analysis and reporting.

- Business analysis
- Report generation
- Decision making
- Example: A retail company stores sales and customer data in a data warehouse to analyze business performance.